UML Diagrams & System Architecture

Enterprise Clean Architecture, State Machine Specifications, Sequence Flows & Entity Data Models • Android Jetpack, Vercel PWA & Hardware Keystore • FNB AOTY 2026 Final MVP

1. Architectural Overview & Clean Architecture

Umkhuseli follows the official Android Enterprise Clean Architecture standard, structured into strictly decoupled layers: **UI (Presentation)**, **Domain / ViewModel**, **Dependency Injection (AppContainer)**, and **Data / Hardware Services**. To ensure zero overhead, deterministic initialization, and instant 0ms memory scrubbing, the application uses a **Manual DI Container** (AppContainer) instead of heavyweight reflection-based injection frameworks. In addition, the cross-platform Web PWA mirrors this architecture via React 19, TypeScript, and Dexter/IndexedDB, hosted on **Vercel Edge**.

Architecture Layer	Primary Components	Key Responsibilities & Technologies
Presentation / UI Layer	MainActivity MunicipalPrepaidCalculator LauncherFacadeScreen SafeHavenScreen	Declarative Jetpack Compose UI & React 19; renders PowerCalc SA facade, store-themed launcher, biometric Safe Haven, and UAT intercept banner.
Domain & ViewModel Layer	SafetyTriggerViewModel SankofaViewModel UatFeedbackManager	Manages UI StateFlow streams; orchestrates 3-second SOS grace period countdown, duress PIN flows, incident sorting, and UAT feedback sync.
Dependency Injection	AppContainer SankofaApplication	Singleton manual DI container; provides thread-safe access to Room DB, AndroidKeyStore crypto helpers, and dispatch services.
Data & Persistence Layer	AppDatabase IncidentDao GuardianDao UserProfileDao SupabaseSync	Room SQLite persistence on Android; IndexedDB on Web; local-first encrypted incident narratives and UAT feedback outbox synced to Supabase.
Hardware & Security Layer	VaultEncryptionHelper EmergencyService EmergencyDispatchHelper	AndroidKeyStore 256-bit AES-GCM StrongBox keys, BiometricPrompt, LocationManager GPS queries, SmsManager broadcasts, and Vibrator.

2. UML State Machine: Emergency SOS Signal Lifecycle

The emergency dispatch system operates as a finite state machine implemented via SafetySignalState sealed interface. State transitions guarantee tactile feedback on countdown ticks, intercept handling under UAT Mode, and atomic persistence to Room DB:

Source State	Transition Event	Target State	System Actions & Side Effects
[Idle]	armQuickAccessSos(3)	**[Armed(sec=3)]**	Launches countdown coroutine; triggers initial haptic vibration pulse.
[Armed(sec>0)]	1-second tick (sec ≥ 1)	**[Armed(sec=sec-1)]**	Updates countdown state; pulses vibrator for tactile confirmation.
[Armed]	cancelArmedSos()	**[Cancelled]**	Cancels coroutine; prevents SMS dispatch; returns to idle upon user acknowledgement.
[Armed]	Countdown reaches 0 (Normal)	**[Dispatched]**	Fetches GPS coordinates; persists IncidentEntity; broadcasts 72-hr beacon SMS to 5 guardians.

[Armed]	Countdown reaches 0 (UAT Mode)	**[SimulatedDispatch]**	Intercepts dispatch to simulated guardians (+27 82 000 0001/2); logs test latency; prompts user feedback.

3. UML Sequence Flows

Sequence 1: Safe Haven Entry & Biometric Decryption

Actor: Survivor • **Components:** PowerCalc SA → SankofaViewModel → BiometricPrompt → VaultEncryptionHelper → Room DB

1. Survivor inputs PIN 1234= or long-presses STS Token button on PowerCalc keypad.
2. MunicipalPrepaidCalculator notifies SankofaViewModel without rendering audio/visual alert.
3. BiometricPrompt prompts biometric challenge; user provides verified fingerprint.
4. VaultEncryptionHelper accesses sankofa_vault_master_key from AndroidKeyStore.
5. IncidentDao.getAllIncidents() queries encrypted ciphertexts from Room SQLite.
6. VaultEncryptionHelper.decrypt() initializes AES-256-GCM cipher with record's 12-byte IV.
7. Decrypted victim narratives and J88 clinical notes are rendered inside SafeHavenScreen.

Sequence 2: 0-Millisecond Panic Camouflage Scrub

Actor: Survivor / Approaching Intruder • **Components:** Back Key / Esc → BackHandler → SankofaViewModel → RAM → UI

1. Intruder approaches; survivor presses hardware Back button, Escape key, or Quick-Exit.
2. Android BackHandler / Web event listener intercepts event synchronously (0 ms latency).
3. SankofaViewModel.resetToCamouflage() zeroes plaintext memory references and resets UI StateFlow to Idle.
4. UI recomposes directly to PowerCalc SA (municipal prepaid electricity calculator).
5. System task thumbnail is sanitized, leaving zero visual traces in the Recent Apps switcher.

4. UML Class Diagram: Data Entities & DAO Relationships

Entity / Class	Key Fields & Types	Statutory / Legal Purpose & Methods
IncidentEntity (Room Table: *incidents*)	id: Long timestamp: Long incidentType: String severityLevel: Int (1-5) perpetrator: String policeObNumber: String description: String (ENC) injuriesOrImpact: String (ENC)	Court-admissible incident log compliant with Domestic Violence Act Form 2 affidavit requirements. Sensitive narrative fields encrypted via AES-256-GCM.
GuardianEntity (Room Table: *guardians*)	id: Long name: String surname: String mobileNumber1: String hasWhatsAppAccess: Boolean isPrimary: Boolean instagramHandle: String tiktokHandle: String	Trusted Emergency ICE Guardian registry. Stores primary/secondary contacts and social handles for multi-channel live beacon alerts.
UserProfileEntity (Room Table: *user_profile*)	id: Long = 1L fullLegalNames: String idOrPassportNumber: String safeContactNumber: String safeServiceAddress: String isAddressWithheld: Boolean digitalSignatureBase64: String duressPin: String = '9999'	Section 4 Domestic Violence Act Complainant Profile. Stores statutory legal particulars and Section 4(2) residential address withholding toggle for court protection.
UatFeedbackEntity (Supabase Table: *uat_feedback_submissions*)	id: UUID session_id: String persona_type: String sos_trigger_rating: Int camouflage_rating: Int overall_rating: Int qualitative_feedback: Text network_type: String	POPIA-compliant anonymous field testing telemetry and user feedback store, with local offline outbox fallback.